OAuth and OpenID Connect

(IN PLAIN ENGLISH)

NATE BARBETTINI
@NBARBETTINI
@OKTADEV

A lot of confusion around OAuth.

- Terminology and jargon
- Incorrect advice

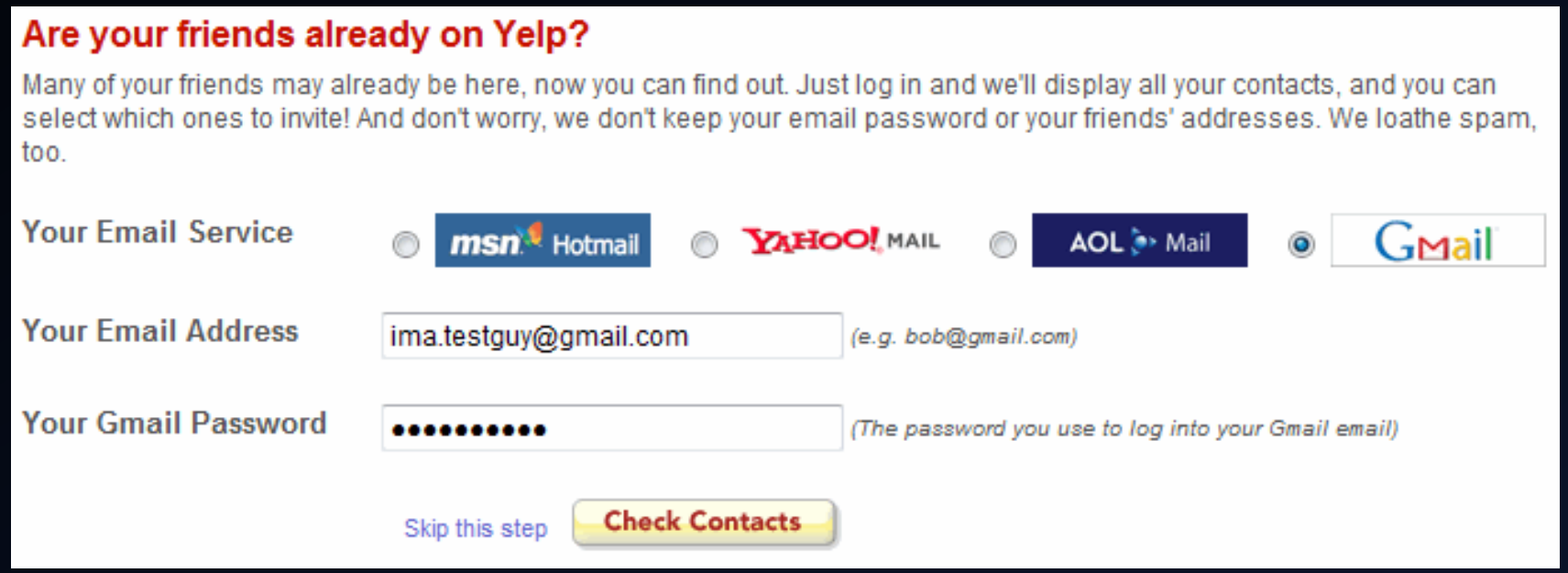
Identity use cases (circa 2007)

- Simple login – forms and cookies
- Single sign-on across sites – SAML
- Mobile app login – ???
- Delegated authorization – ???

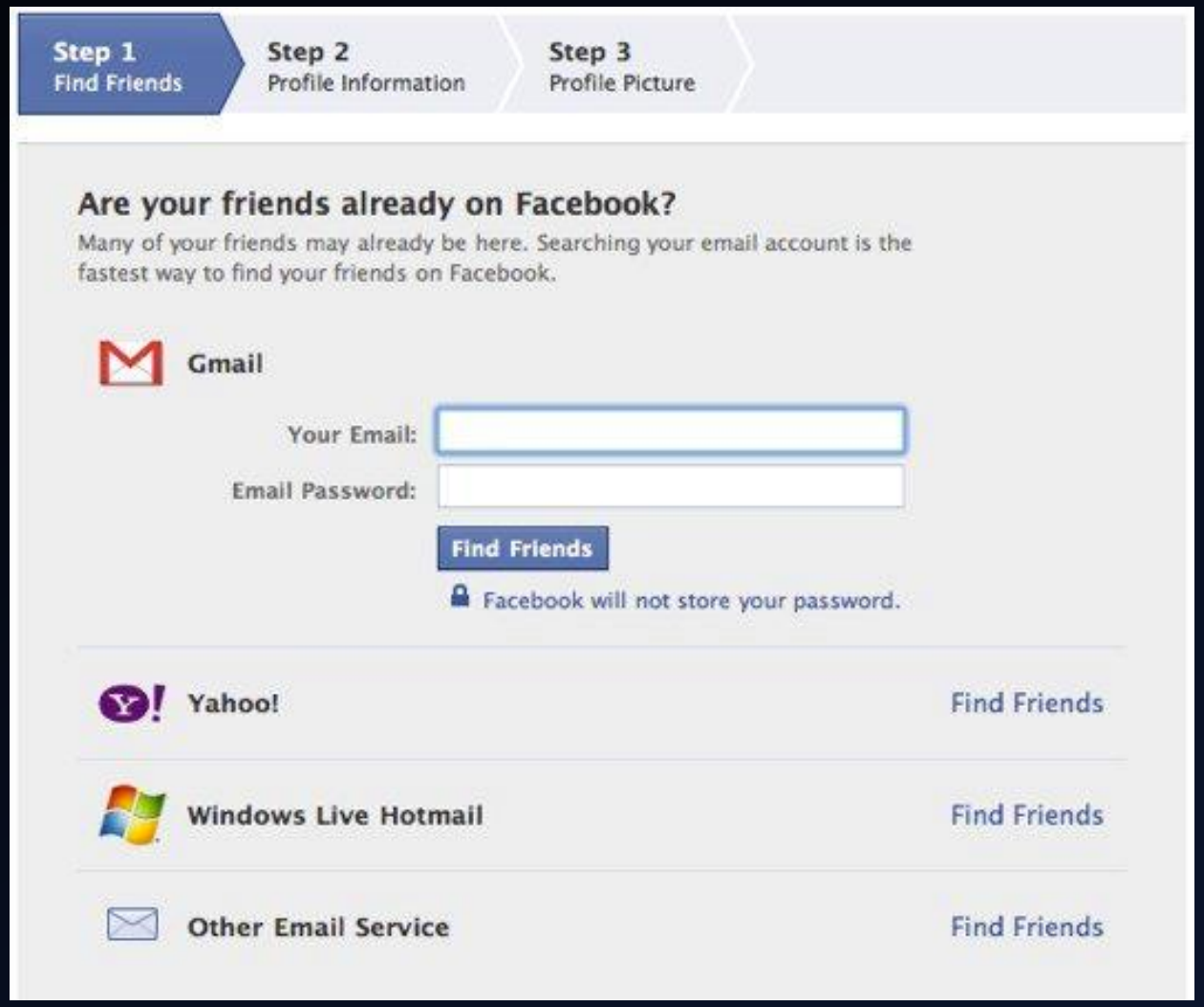
The delegated authorization problem

HOW CAN I LET A WEBSITE ACCESS MY DATA (WITHOUT GIVING IT MY PASSWORD)?

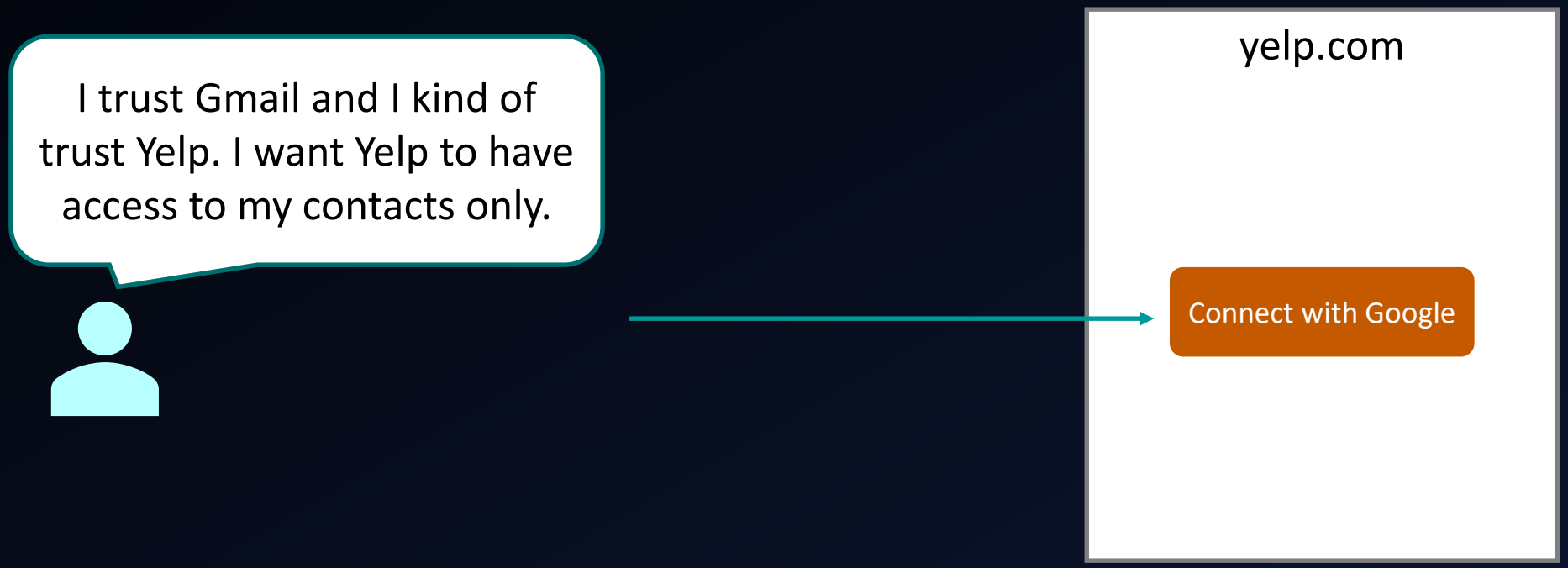
Don't do it this way!



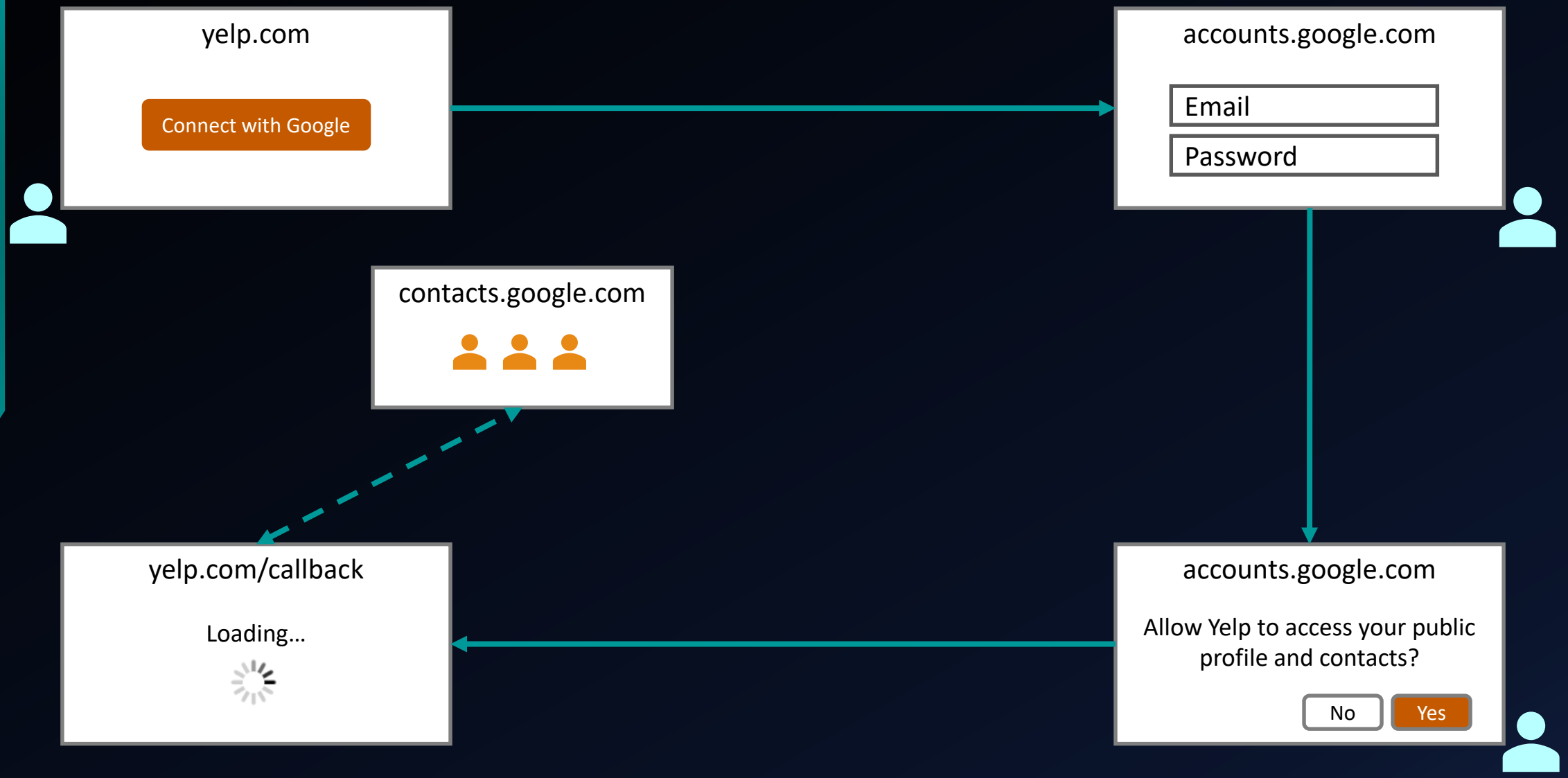
Don't do it this way!



Delegated authorization with OAuth 2.0



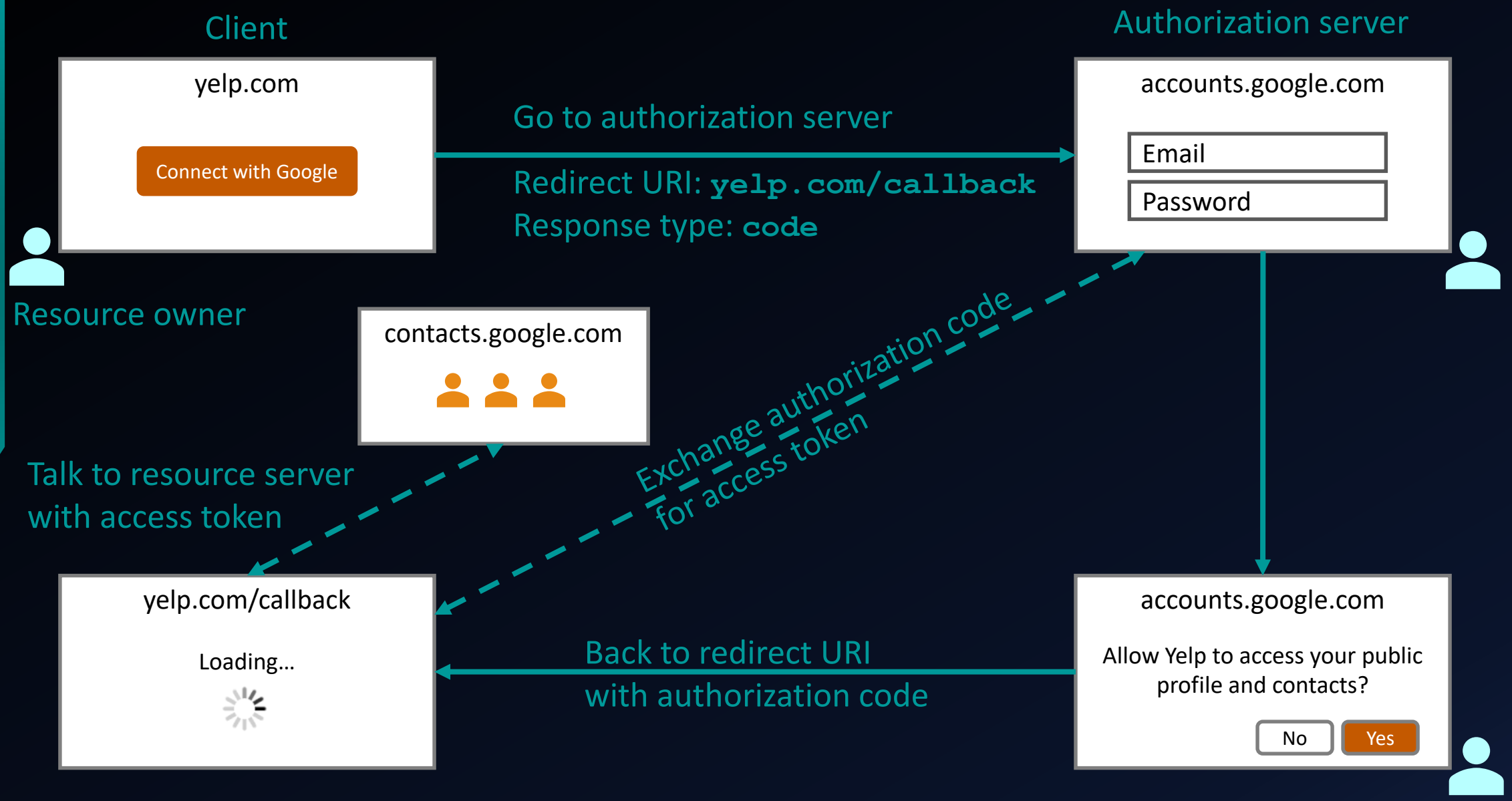
Delegated authorization with OAuth 2.0



OAuth 2.0 terminology

- Resource owner
- Client
- Authorization server
- Resource server
- Authorization grant
- Access token

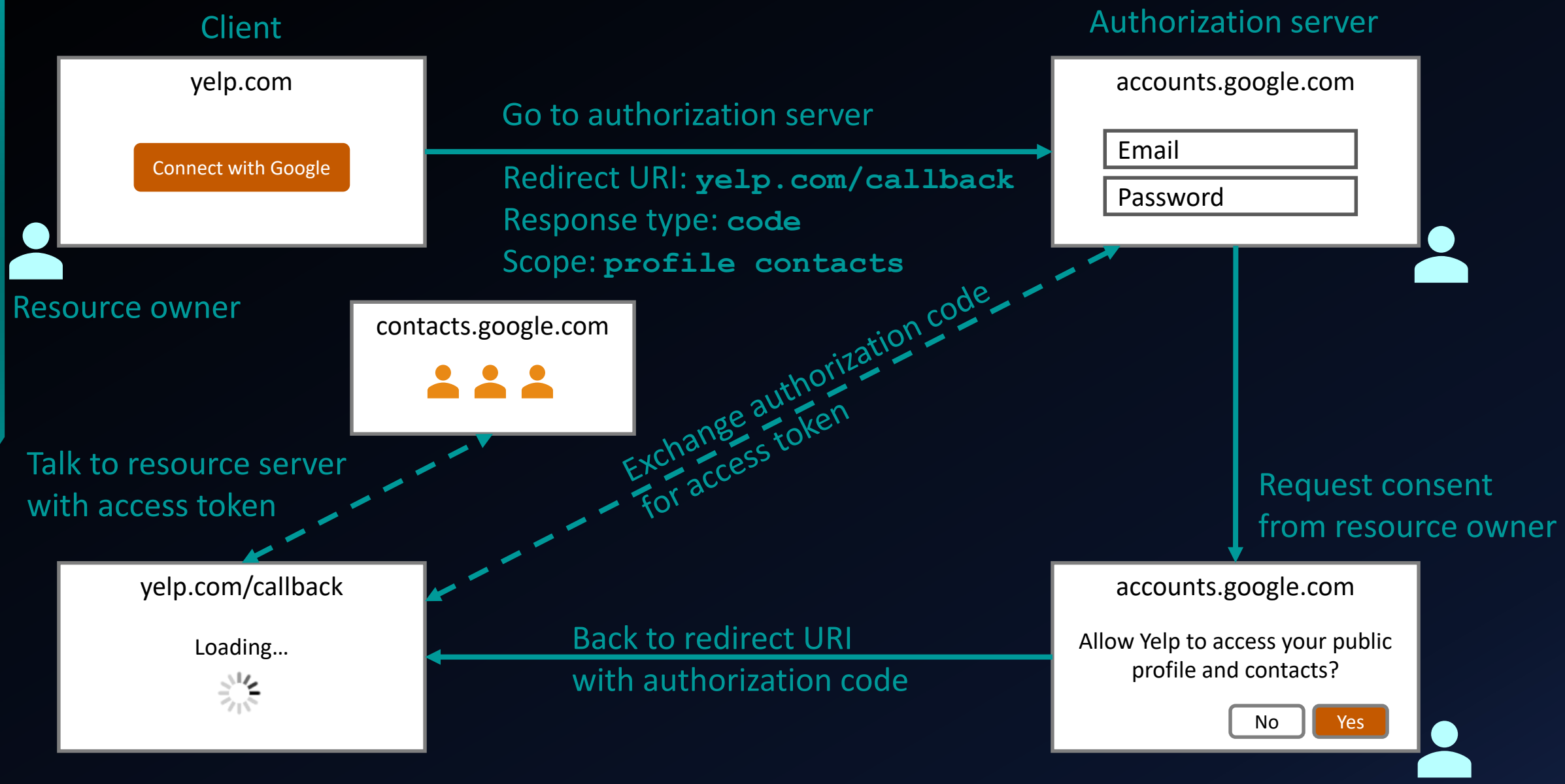
OAuth 2.0 authorization code flow



More OAuth 2.0 terminology

- Scope
- Consent

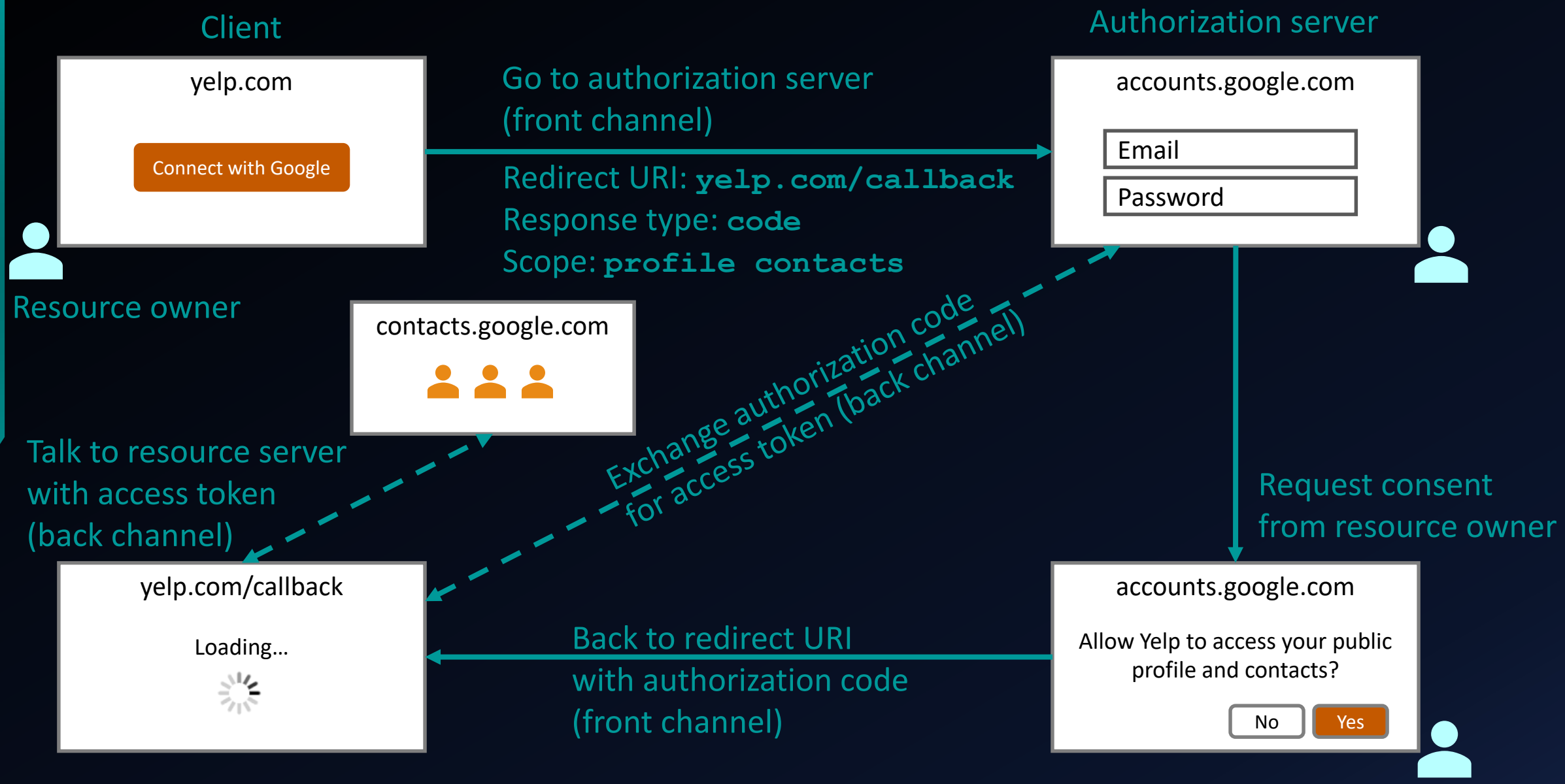
OAuth 2.0 authorization code flow



Even more OAuth 2.0 terminology

- Back channel (highly secure channel)
- Front channel (less secure channel)

OAuth 2.0 authorization code flow



Starting the flow

```
https://accounts.google.com/o/oauth2/v2/auth?
 client_id=abc123&
 redirect_uri=https://yelp.com/callback&
 scope=profile&
 response_type=code&
 state=foobar
```

Calling back

```
https://yelp.com/callback?
 error=access_denied&
 error_description=The user did not consent.
```

```
https://yelp.com/callback?
 code=oMsCeLvIaQm6bTrgtp7&
 state=foobar
```

Exchange code for an access token

```
POST www.googleapis.com/oauth2/v4/token
Content-Type: application/x-www-form-urlencoded

code=oMsCeLvIaQm6bTrgtp7&
client_id=abc123&
client_secret=secret123&
grant_type=authorization_code
```

Authorization server returns an access token

```
{
  "access_token": "fFAGRNJru1FTz70BzhT3Zg",
  "expires_in": 3920,
  "token_type": "Bearer",
}
```

Use the access token

```
GET api.google.com/some/endpoint
Authorization: Bearer fFAGRNJru1FTz70BzhT3Zg
```

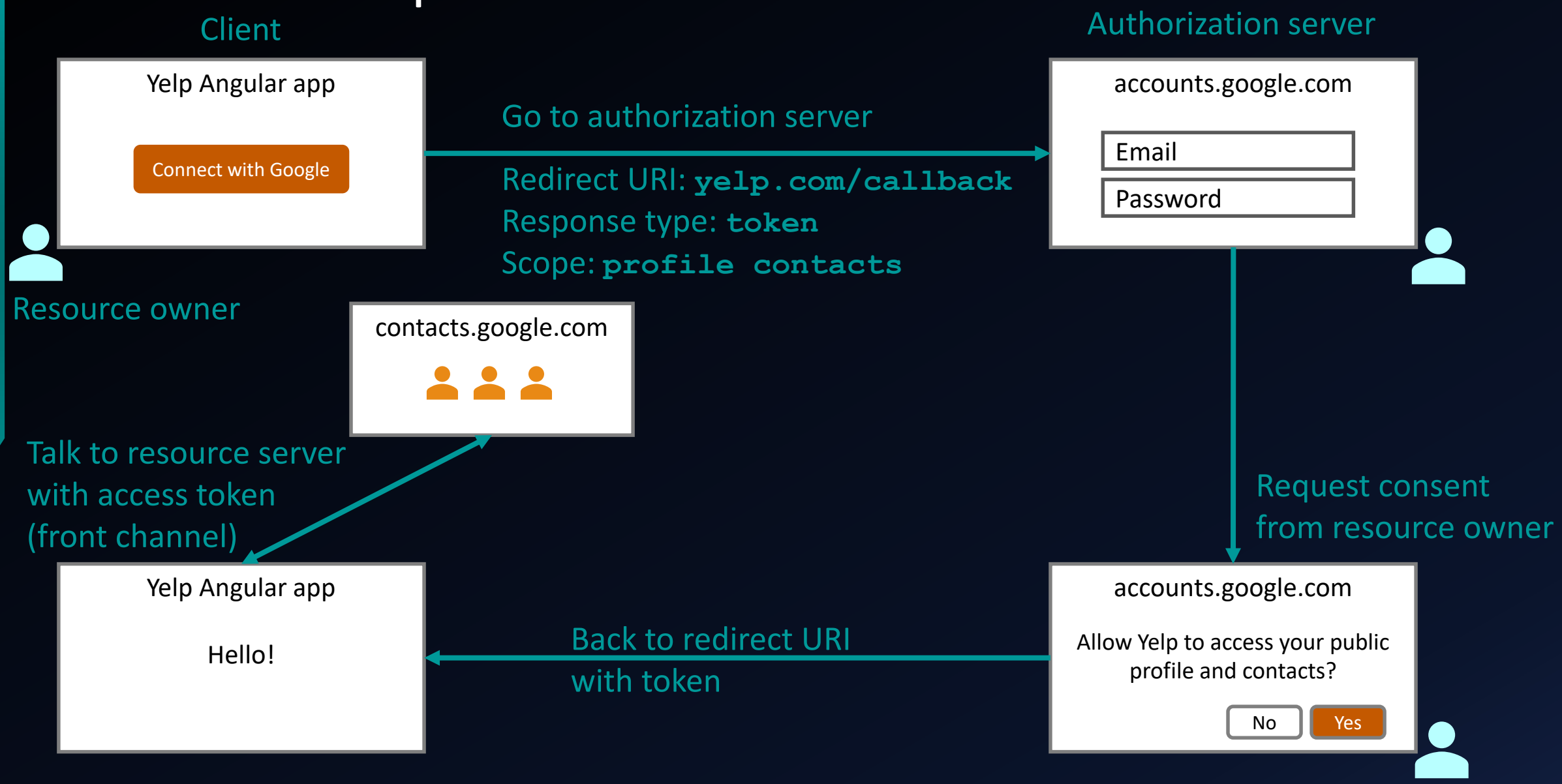
- Validate token
- Use token scope for authorization

OAuth 2.0 flows

- Authorization code (front channel + back channel)
- Implicit (front channel only)

- Resource owner password credentials (back channel only)
- Client credentials (back channel only)

OAuth 2.0 implicit flow



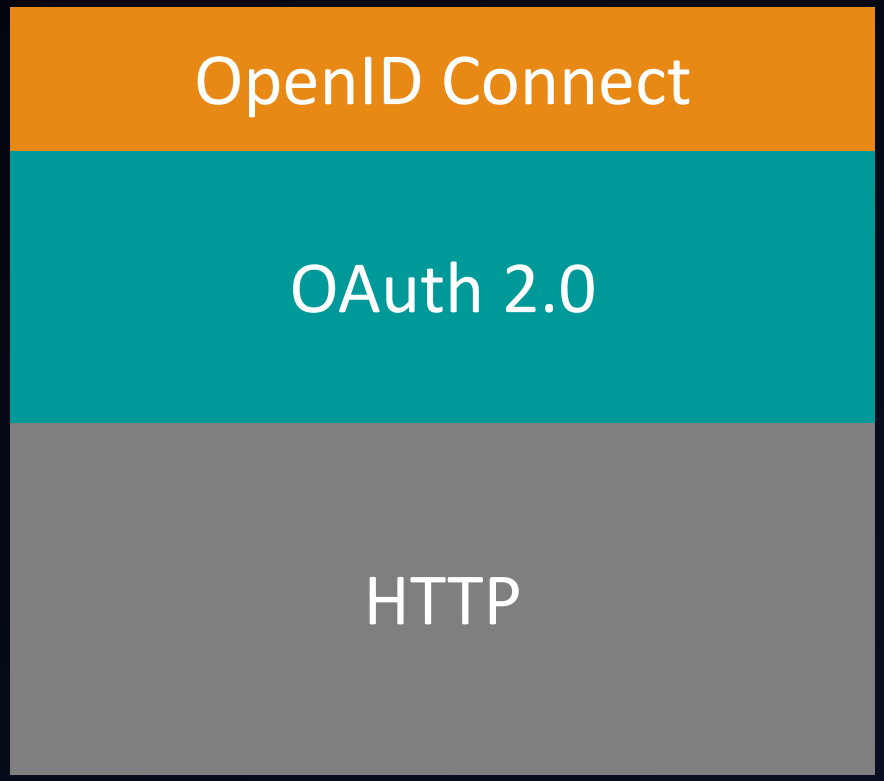
Identity use cases (circa 2012)

• Simple login – OAuth 2.0	Authentication
• Single sign-on across sites – OAuth 2.0	Authentication
• Mobile app login – OAuth 2.0	Authentication
• Delegated authorization – OAuth 2.0	Authorization

Problems with OAuth 2.0 for **authentication**

- No standard way to get the user's information
- Every implementation is a little different
- No common set of scopes

OAuth 2.0 and OpenID Connect

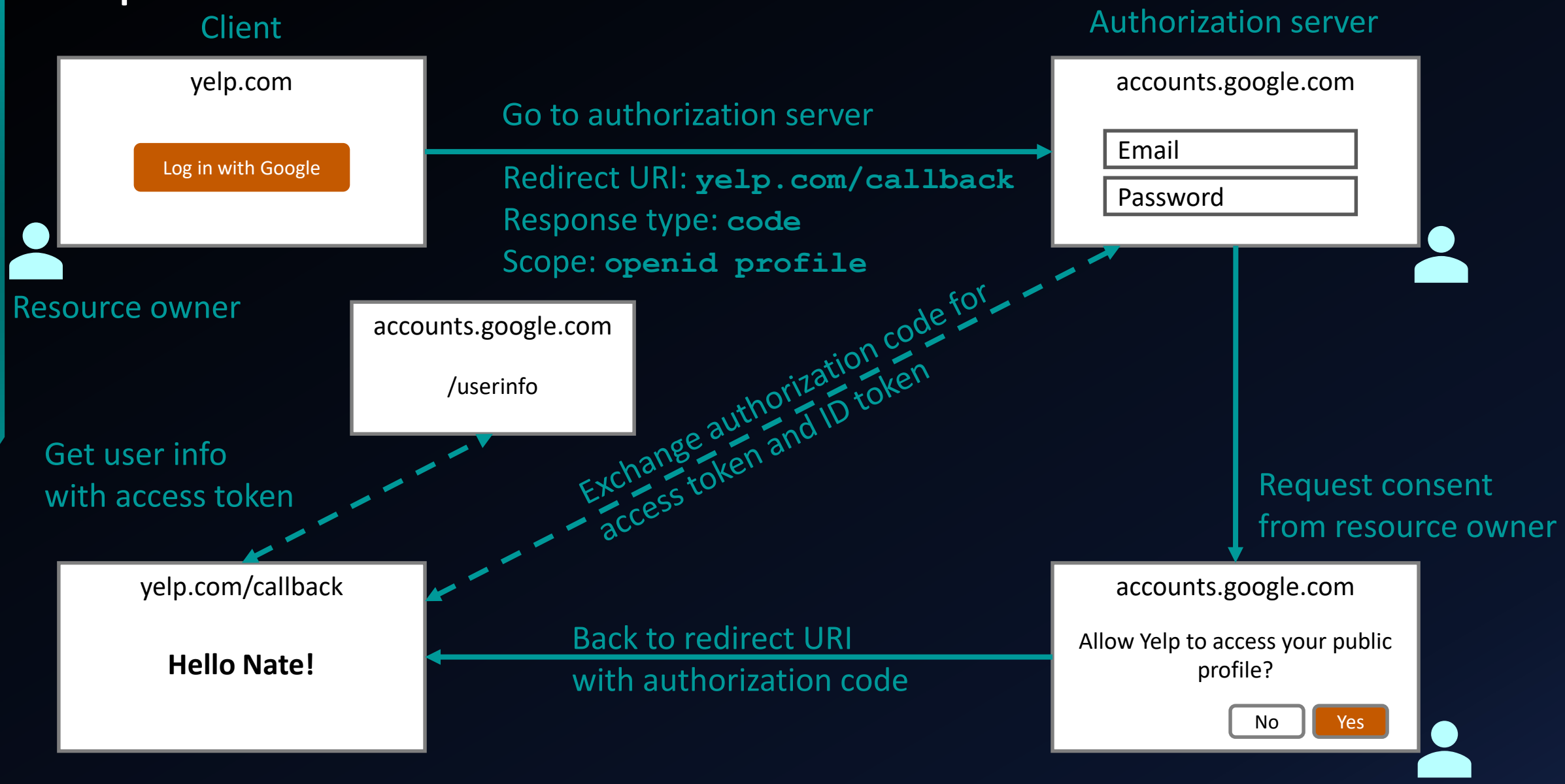


- OpenID Connect is for **authentication**
- OAuth 2.0 is for **authorization**

What OpenID Connect adds

- ID token
- UserInfo endpoint for getting more user information
- Standard set of scopes
- Standardized implementation

OpenID Connect authorization code flow



Starting the flow

```
https://accounts.google.com/o/oauth2/v2/auth?
 client_id=abc123&
 redirect_uri=https://yelp.com/callback&
 scope=openid profile&
 response_type=code&
 state=foobar
```

Exchange code for access token **and** ID token

```
POST www.googleapis.com/oauth2/v4/token
Content-Type: application/x-www-form-urlencoded

code=oMsCeLvIaQm6bTrgtp7&
client_id=abc123&
client_secret=secret123&
grant_type=authorization_code
```

Authorization server returns access and ID tokens

```
{
  "access_token": "fFAGRNJru1FTz70BzhT3Zg",
  "id_token": "eyJraB03ds3F..."
  "expires_in": 3920,
  "token_type": "Bearer",
}
```

ID token (JWT)

eyJhbGciOiJSUzI1NiIsImtpZCI6IkRNa3Itd0JqRU1EYnhOY25xaVJISVhu
YUxubWI3UUpfWF9rWmJyaEtBMGMifQ
.
eyJzdWIiOiIwMHU5bzFuaWtqdk9CZzVabzBoNyIsInZlciI6MSwiaXNzIjoi
aHR0cHM6Ly9kZXYtMzQxNjA3Lm9rdGFwcmV2aWV3LmNvbS9vYXV0aDIvYXVz
OW84d3ZraG9ja3c5VEwwaDciLCJhdWQiOiJsWFNlbkx4eFBpOGtRVmpKRTVz
NCIsImlhdCI6MTUwOTA0OTg5OCwiZXhwIjoxNTA5MDUzNDk4LCJqdGkiOiJJ
RC5oa2RXSXNBSXZTbnBGYVFHTVRYUGNVSmhhMkgwS2c5Ykl3ZEVvVm1ZZHN3
IiwiYW1yIjpbImtiYSIsIm1mYSIsInB3ZCJdLCJpZHAiOiIwMG85bzFuaWpr
aWpLeGNpbjBoNyIsIm5vbmNlIjoidWpwMmFzeHlqN2UiLCJhdXRoX3RpbWUi
OjE1MDkwNDk3MTl9
.
dv4Ek8B4BDee1PcQT_4zm7kxDEY1sRIGbLoNtlodZcSzHz-
XU5GkKyl6sAVmdXOIPUlAIrJAhNfQWQ-
_XZLBVPjETiZE8CgNg5uqNmeXMUnYnQmvN5oWlXUZ8Gcub-GAbJ8-
NQuyBmyec1j3gmGzX3wemke8NkuI6SX2L4Wj1PyvkknBtbjfiF9ud1-
ERKbobaFbnjDFOFTzvL6g34SpMmZWy6uc_Hs--n4IC-ex-
Ps3FcMwRggCW-7o2FpH6rJTOGPZYrOx44n3ZwAu2dGm6axtPI-
sqU8b6sw7DaHpogD_hxsXgMIOzOBMbYsQEiczoGn71ZFz_1O7FiW4dH6g

Header

Payload (claims)

Signature

The ID token (JWT)

```
(Header)
.
{
   "iss": "https://accounts.google.com",
   "sub": "you@gmail.com",
   "name": "Nate Barbettini"
   "aud": "s6BhdRkqt3",
   "exp": 1311281970,
   "iat": 1311280970,
   "auth_time": 1311280969,
}
.
(Signature)
```

Calling the userinfo endpoint

```
GET www.googleapis.com/oauth2/v4/userinfo
Authorization: Bearer fFAGRNJru1FTz70BzhT3Zg

200 OK
Content-Type: application/json

{
    "sub": "you@gmail.com",
    "name": "Nate Barbettini"
    "profile_picture": "http://plus.g.co/123"
}
```

Identity use cases (today)

• Simple login – OpenID Connect	Authentication
• Single sign-on across sites – OpenID Connect	Authentication
• Mobile app login – OpenID Connect	Authentication
• Delegated authorization – OAuth 2.0	Authorization

OAuth and OpenID Connect

Use OAuth 2.0 for:

- Granting access to your API
- Getting access to user data in other systems

(Authorization)

Use OpenID Connect for:

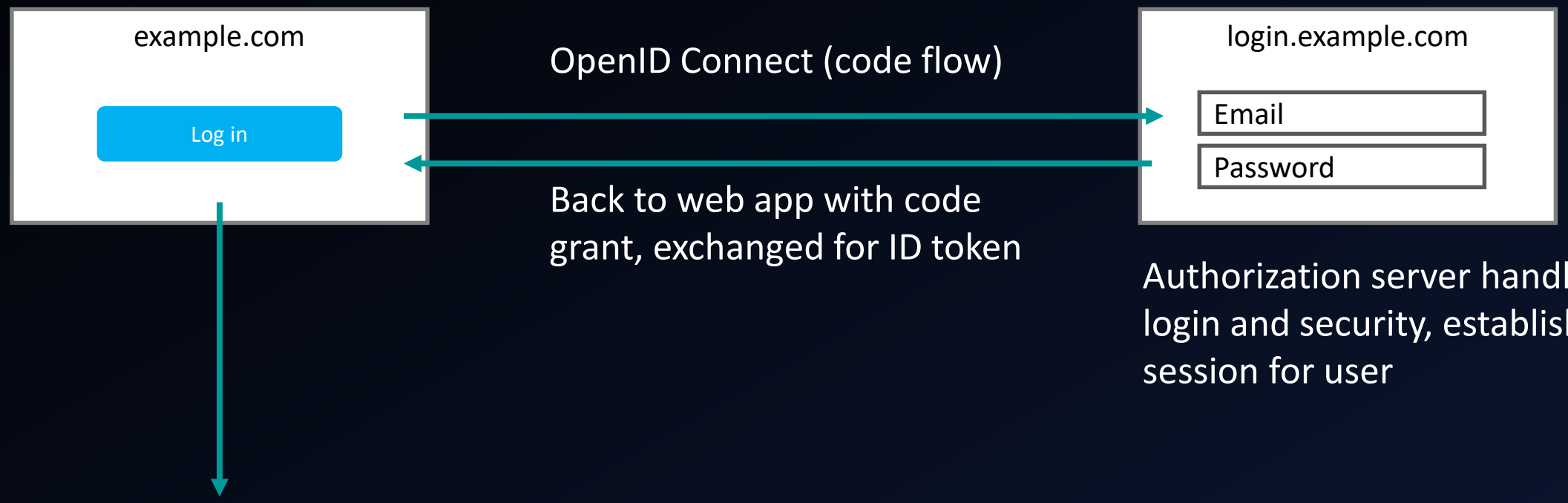
- Logging the user in
- Making your accounts available in other systems

(Authentication)

Which flow (grant type) do I use?

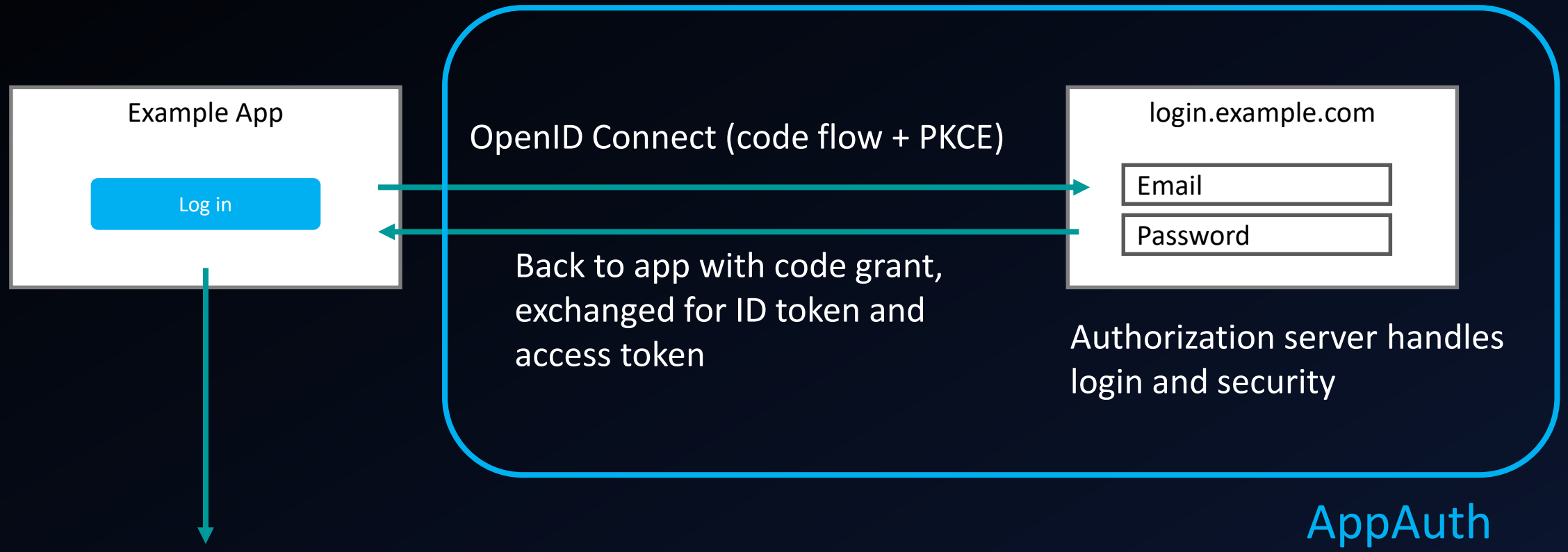
- Web application w/ server backend: **authorization code flow**
- Native mobile app: **authorization code flow with PKCE**
- JavaScript app (SPA) w/ API backend: **implicit flow**
- Microservices and APIs: **client credentials flow**

Example: web application with server backend



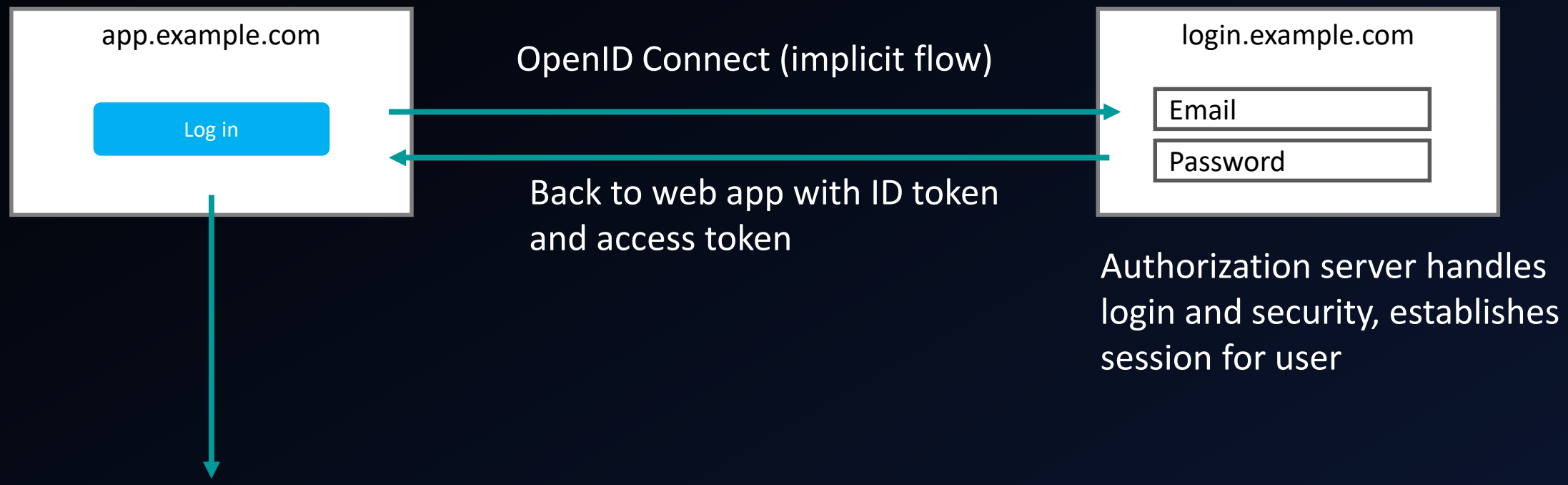

```
Set-Cookie: sessionid=f00b4r; Max-Age: 86400;
```

Example: native mobile app



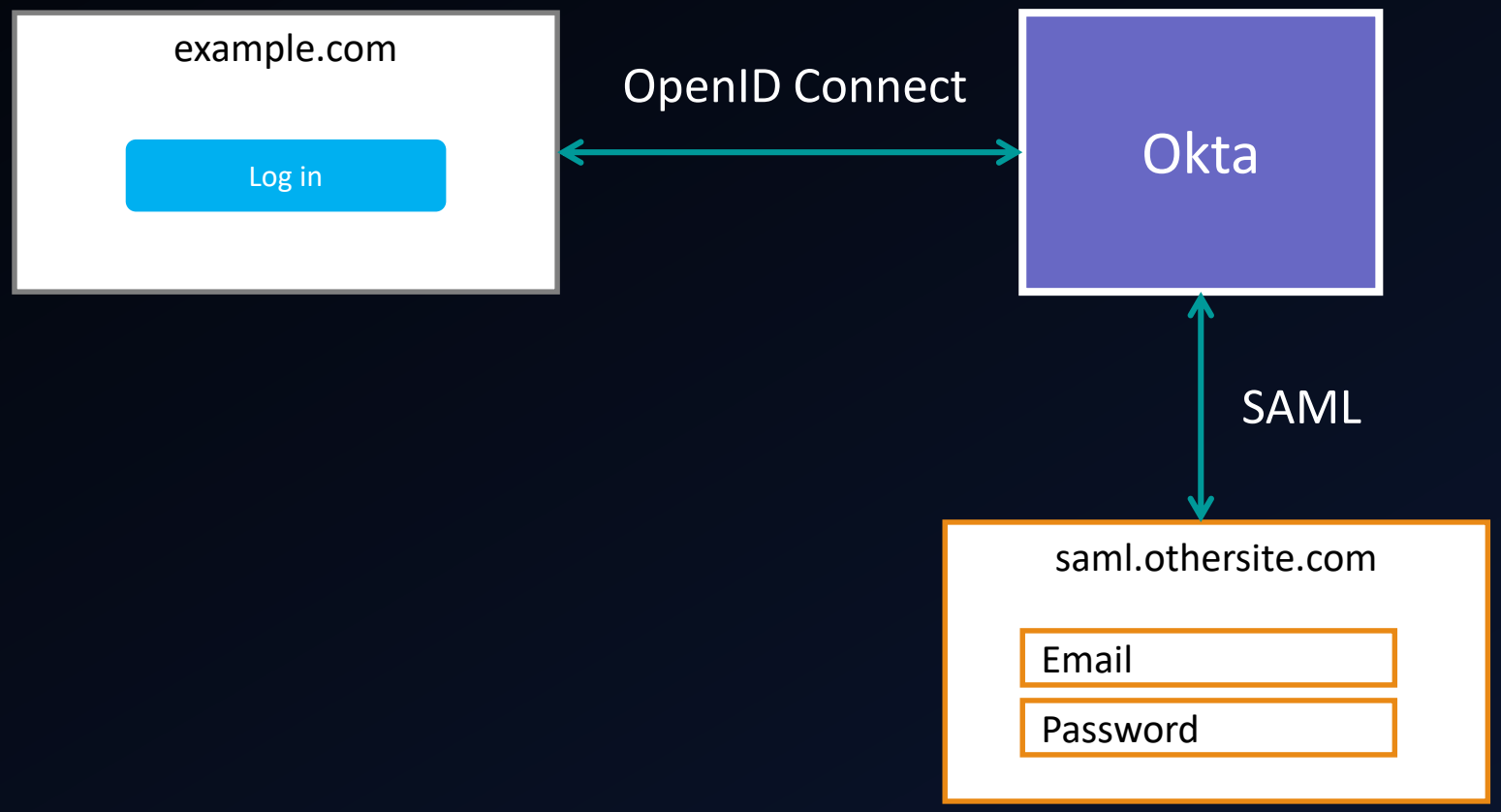
Store tokens in protected device storage
Use ID token to know who the user is
Attach access token to outgoing API requests

Example: SPA with API backend



Store tokens locally with JavaScript
Use ID token to know who the user is
Attach access token to outgoing API requests

Example: SSO with 3rd-party services



Token validation

- The fast way: local validation
  - Check expiration timestamp
  - Validate cryptographic signature
- The strong way: introspection

Revocation

Token issued and used for API calls

Device compromised!

What happens?

12PM

1PM

2PM

```
POST /oauth2/default/v1/revoke
Content-Type: application/x-www-form-urlencoded

token=fFAGRNJru1FTz70BzhT3Zg
&token_type_hint=access_token
&client_id=...
```

Keeping the user signed in

For both local validation and introspection, the token is invalid once it expires, so:

- If there's a user at the keyboard, just redirect through the authorization server again.
- If there's no user (automated tasks), request a refresh token (offline scope).

Thanks y'all!

Nate Barbettini
@nbarbettini

oauth.com
@oktadev

Free hosted authorization server:
developer.okta.com